

VIBRATION

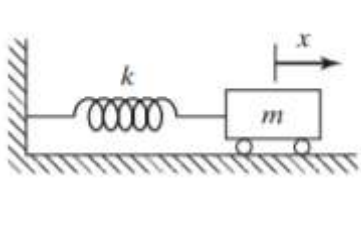
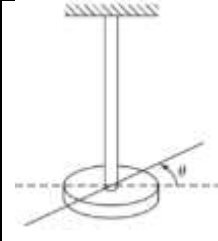
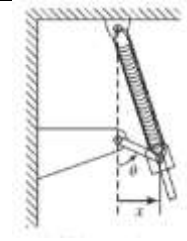
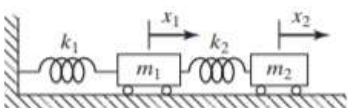
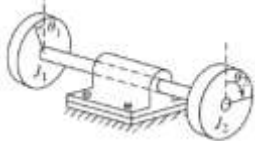
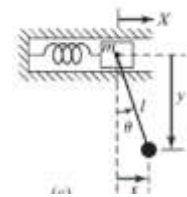
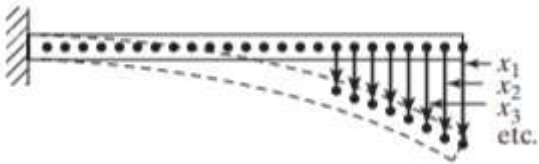
Vibration is a repetitive, periodic, oscillatory response of a mechanical system. Let us consider a simple example of a simple pendulum. It is pulled to one side and released. It oscillates. When it is at the extreme positions, it has got potential energy. As it reaches the equilibrium position, it has only kinetic energy. After some time, it loses energy and comes to rest. These processes can be seen in any vibrating system. That is, every vibrating system will have three basic elements

1. Kinetic energy storing elements (inertia elements)
2. Potential energy/strain energy storing elements (stiffness elements/spring elements)
3. Energy dissipating elements (damping elements)

Hence, any vibratory system can be represented by a mass-spring-damper.

Degree of freedom

The minimum number of independent co-ordinates required to determine completely the positions of all parts of the system is known as the **degrees of freedom**

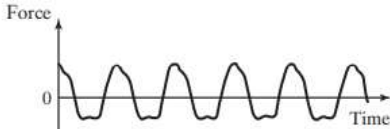
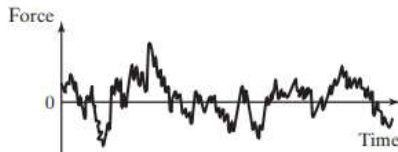
Single degree of freedom			
Two degrees of freedom			
Infinite degrees of freedom			

The number of degrees of freedom is equal to the number of mass elements in the system.

Discrete and Continuous system: If a vibrating system can be represented by finite number of mass elements (however large that number may be), it is considered as a discrete system. It will have finite number of natural frequencies equal to the number of mass elements. Such systems will be governed by the **ordinary differential equations**.

Where as, the continuous system will have infinite degrees of freedom and hence infinite number of natural frequencies. Such systems will be governed by **partial differential equations**.

Classification of vibration

1.	Free If the system is disturbed from its equilibrium position and is allowed to vibrate on its own without any external force or help, it is called free vibration. The frequency with which it vibrates freely is known as the natural frequency of the system. The number of natural frequencies of a system is equal to the number of degrees of freedom of the system.	Forced If a system is subjected to an external force continuously or intermittently, the resulting vibration is known as forced vibration. If the frequency of the external force is equal to the natural frequency of the system, then a condition of resonance occurs and the system vibrates with large amplitudes and may cause failure.
2.	Undamped If no energy is lost from the system, it known as undamped vibration. Strictly speaking, no physical vibrating system is undamped. But, in many cases, the damping is so negligible that we can neglect it and consider as undamped.	Damped If energy is dissipated from the system, it is called damped vibration. The energy is lost mainly due to friction and we can classify damping as viscous damping (fluid friction), coulomb damping (dry friction), and material damping (internal molecular friction).
3.	Linear If all the elements of a vibrating system behave linearly, it is known as linear vibration. It will be characterised by linear equations and the principle of superposition will be valid in such systems. The solution methods of such equations are well developed.	Non-linear If any of the components (stiffness, inertia, or damping) behaves non-linearly, the resulting vibration is non-linear vibration. The governing equations will be non-linear and there exists no closed form solutions to such equations. We may need to use numerical methods or piece-wise linearisation to solve such equations. Most of the physical systems behave non-linearly with large amplitudes of oscillations.
4.	Deterministic If the magnitude of excitation of the system at any instant of time can be predicted, the resulting vibration is known as deterministic.  (a) A deterministic (periodic) excitation	Stochastic (Random) If the exciting force at any instant can not be predicted, the vibration becomes random (eg. Vibration due to wind, road undulations, earth quake etc). The vibratory response of the system is also random. It can be represented only with the help of statistical quantities.  (b) A random excitation

Consider the example of a simple pendulum. The equation governing the motion is

$$I_o \ddot{\theta} = \Sigma M_o \gg \gg ml^2 \ddot{\theta} = -mgl \sin \theta$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

This is a non-linear equation because

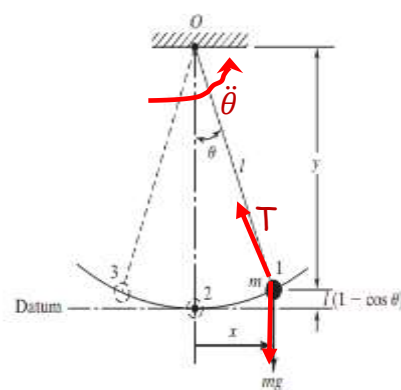
$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \quad (\theta \text{ is in radian})$$

For small angles, $\sin \theta \cong \theta$.

$$\therefore \ddot{\theta} + \frac{g}{l} \theta = 0 \quad \text{or} \quad \ddot{\theta} + \omega^2 \theta = 0$$

This is a linear differential equation whose solution is

$$\theta(t) = A \sin \omega t + B \cos \omega t \text{ and period } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$$



Vibration Analysis Procedure

1. Mathematical Modelling

The purpose of mathematical modelling is to represent all important features of the system in order to obtain the governing equation. The mathematical model may be linear or non-linear depending upon the behavior of the system components. This is the most important step in the analysis. As an engineer, you need to use your expertise and understanding in making the mathematical model. All the subsequent analysis depends on the mathematical model.

2. Derivation of governing differential equation

Once the mathematical model is available, the governing equation can be derived by applying universal principles/laws (Newton's laws of motion, D'Alembert's principle, Conservation of energy principle etc). The governing equations may be ordinary or partial differential equations depending on whether the system is discrete or continuous.

3. Solution of governing equations

The governing equations must be solved to get the response of the system. Depending on the nature of the equation, we can use standard closed form solutions, Laplace transform method, Numerical methods etc.

4. Interpretation of the results

The solution of the governing equation gives displacement, velocity, acceleration etc. This must be interpreted with a clear understanding of the purpose of analysis and design requirements.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

